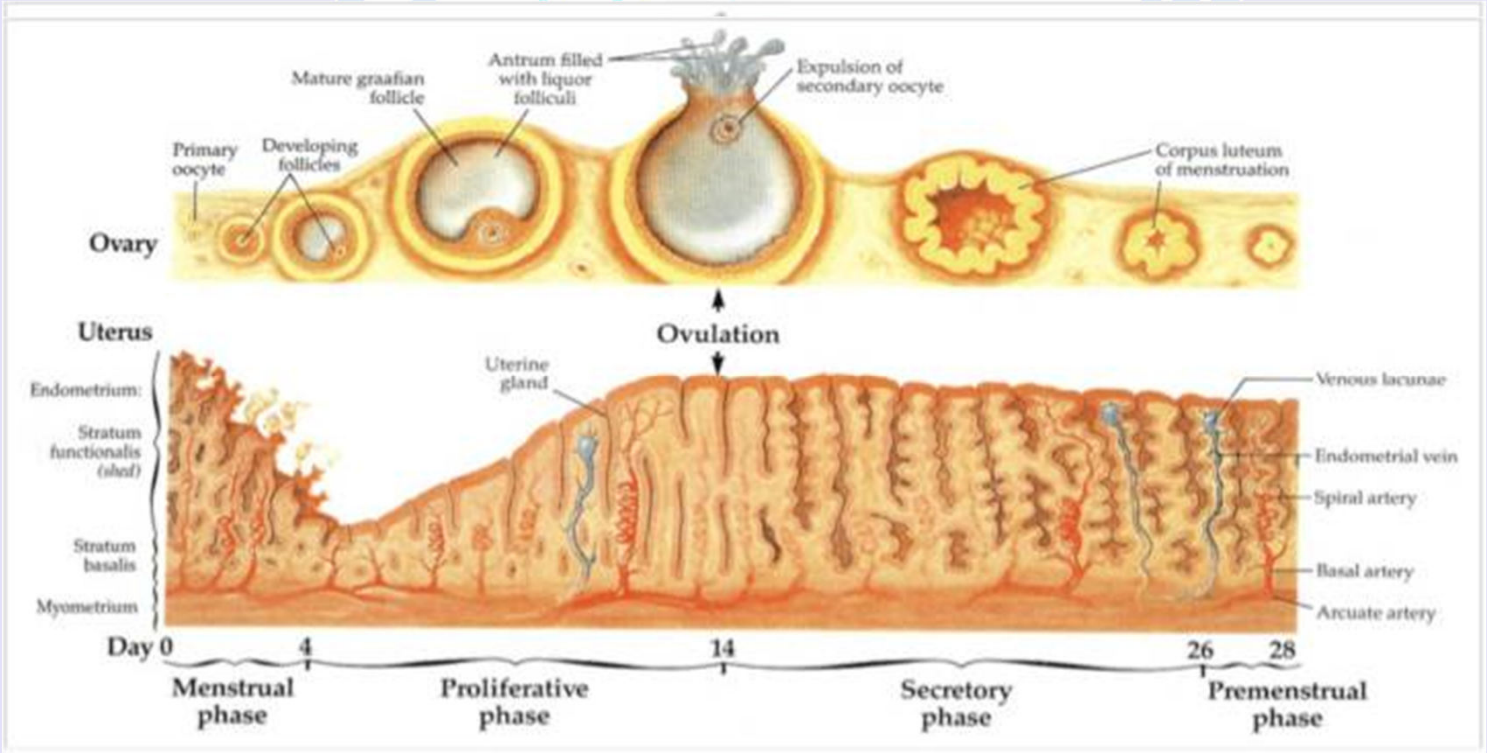
OVARIAN CYCLE

LEARNING OBJECTIVES

- At the end of the lecture,student should be able to :
- Define ovarian cycle
- Explain the stages of maturation of primary oocyte to mature ovum after puberty
- Describe the hormonal control (FSH,LH) of ovarian cycle
- Mention the liberation of mature ovum into abdominal cavity,formation of corpus luteum and albican

OVARIAN CYCLE

- **Follicular Phase (Day 1 to Day 13):**
  - Menstruation lasts from 1 to day 5 of the average cycle.
  - Secretion of Ovarian steroid hormones are at their lowest.



OVARIAN CYCLE

Follicular Phase (Day 1 to Day 13):

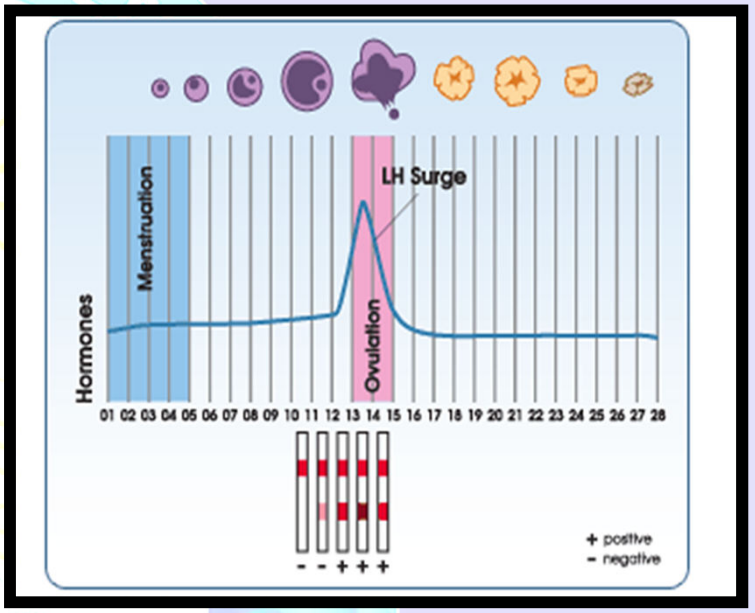
- 12 to 15 primary follicles grow at a time, develop vesicles, and only one reaches to the full maturity remaining are arrested and degenerated.
- Towards the end of follicular phase , one follicle in one ovary reaches maturity and reaches the graffian follicle.
- As follicles grow, the granulosa cells secrete an increasing amount of estradiol (the principal estrogen), which reaches its highest concentration in the blood at about day 12 of the cycle, 2 days before ovulation.

OVARIAN CYCLE

- **Follicular Phase (Day 1 to Day 13):**
  - Growth of follicles and secretion of estradiol are stimulated by FSH secreted by anterior lobe of pitutary gland.
  - FSH secreted in early follicular phase is slightly greater than secreted in late follicular phase.
  - Towards the end of the follicular phase, FSH and estradiol also stimulate the production of LH receptors in thr graffian follicle.
  - Graafian follicle is prepared for the next major event.

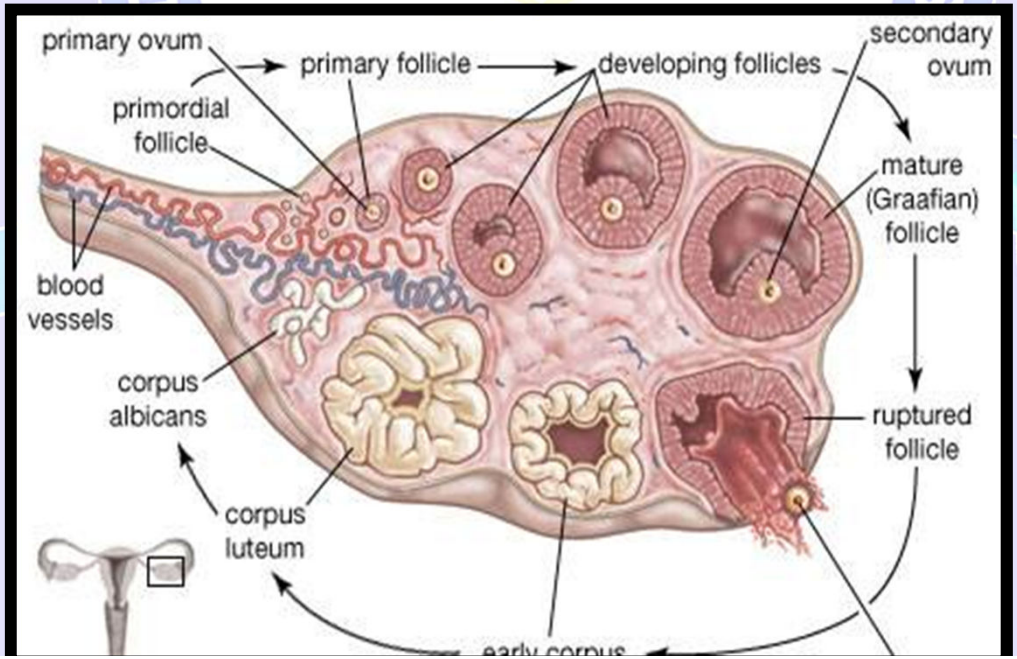
LH SURGE

- Begins about 24 hours before ovulation.
- Reaches its peak 16 hours before ovulation.
- LH surge acts to trigger ovulation.
  - Positive feedback effect of estradiol on the anterior pituitary ,increased LH secretion in late follicular phase and large qauntity of LH culminates blood.



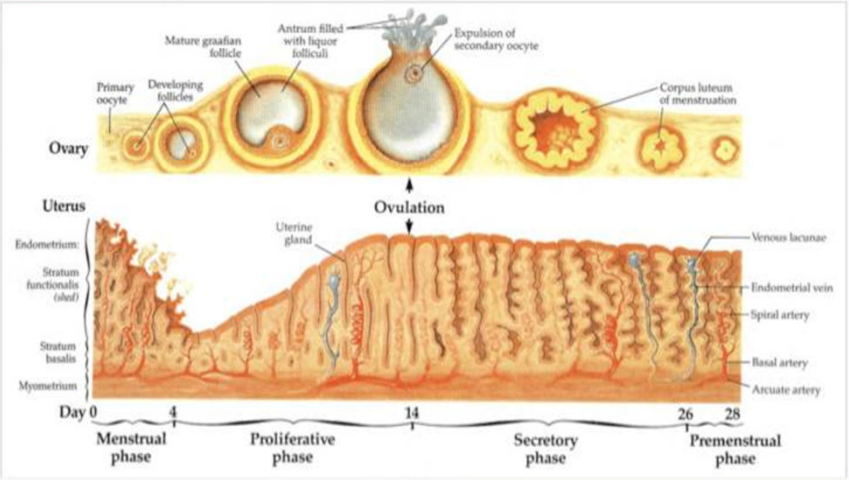
OVULATION

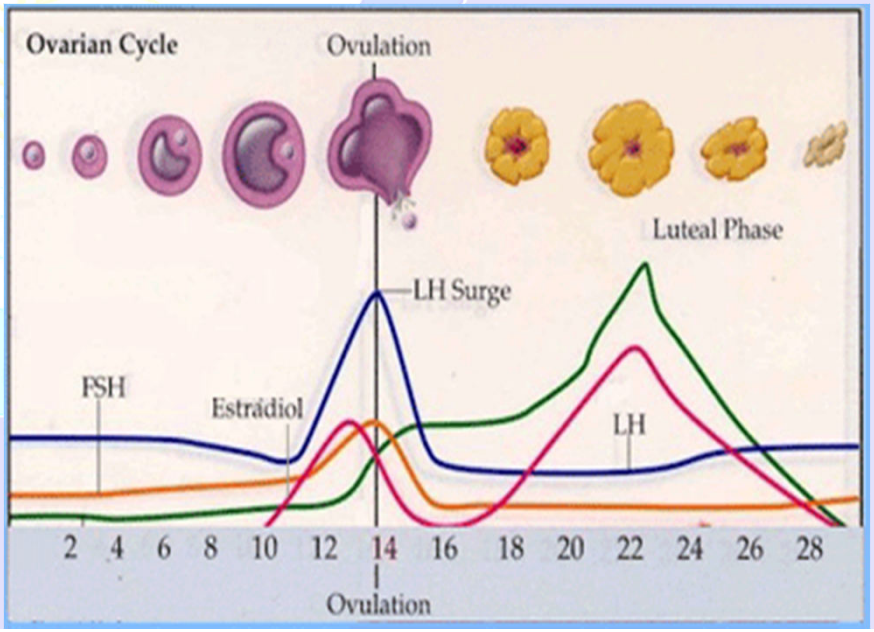
- Mature follicle reaches beneath the surface of ovary producing an elevation, the stigma
- Surge in LH secretion causes the wall of the graafian follicle to rupture at about day 14 and release of mature ovum in the abdominal cavity .



OVULATION

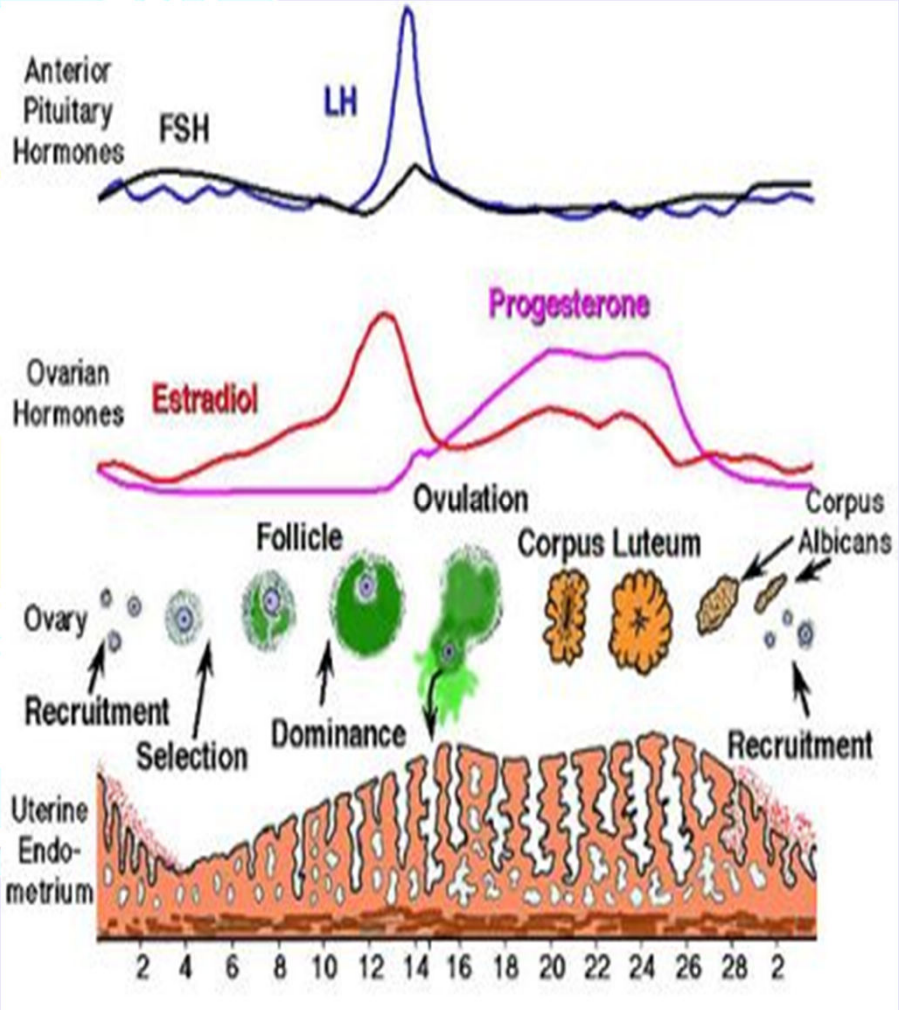
- A secondary oocyte arrested at metaphase
- II of meiosis , is released from the ovary
- and swept by cilia into the uterine tube.
- Ovulation occurs, therefore as a result of
- effects of FSH and LH on the ovarian follicles.





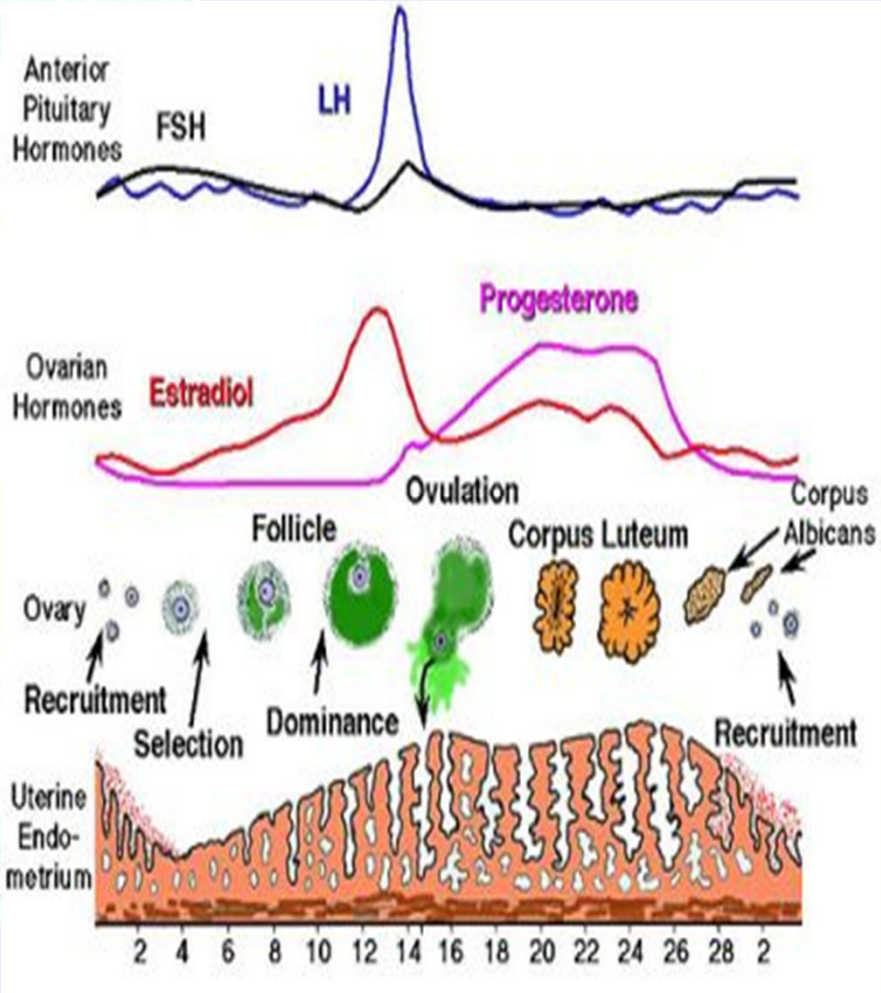
LUTEAL PHASE

- After Ovulation, the empty follicle which remains behind in the ovary is stimulated by LH to become corpus luteum (yellow body).
- Corpus luteum secretes both estradiol and progesterone.
- Progesterone levels in the blood are negligible before ovulation but rise rapidly to a peak level during the luteal phase, approximately one week after ovulation.

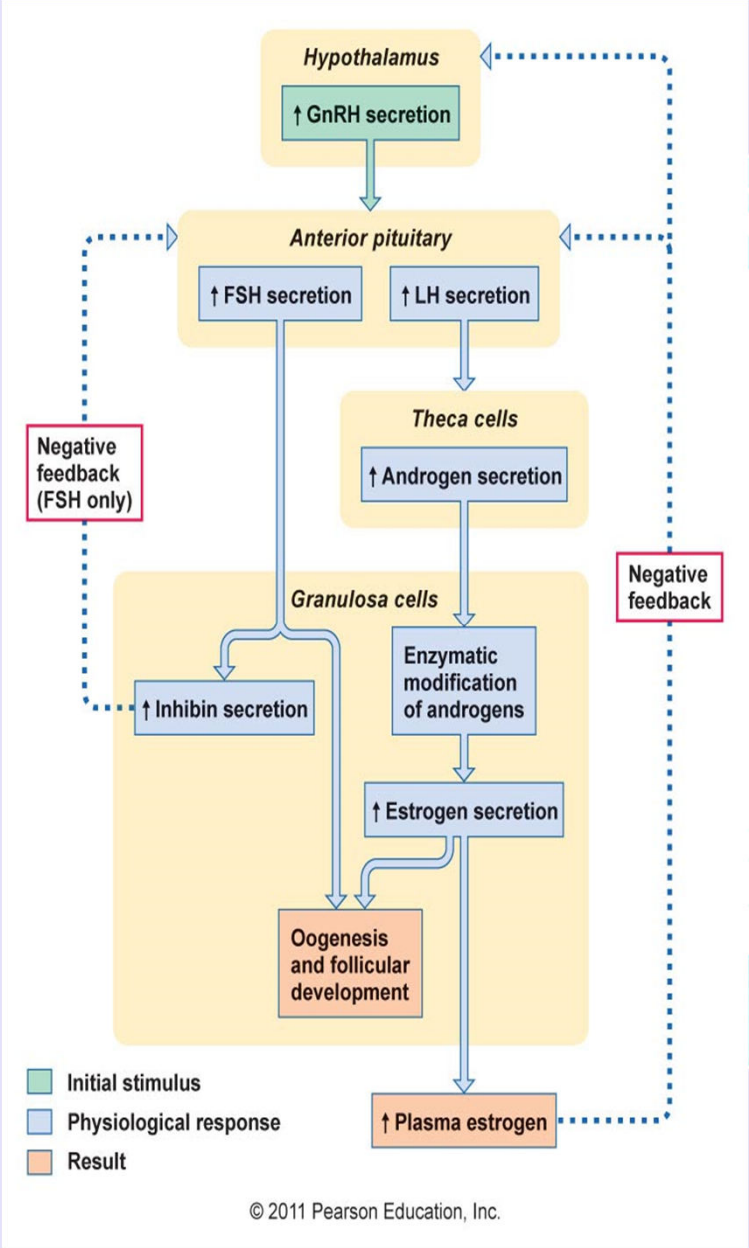


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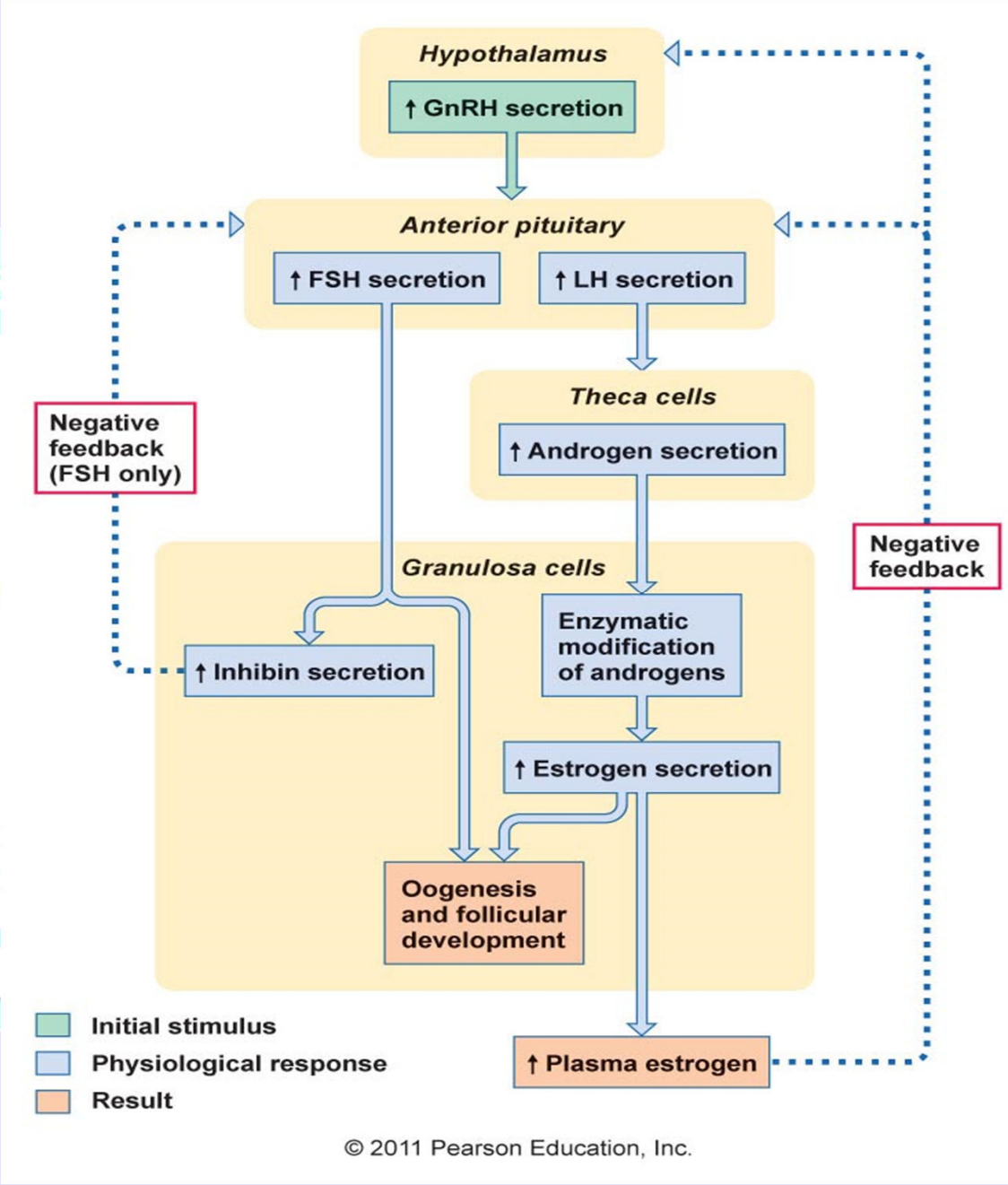
LUTEAL PHASE



- High levels of progesterone combined with estradiol during the luteal phase exert an inhibitory, or negative feedback effect on FSH and LH secretion from anterior pituitary.
- Corpus luteum produces inhibin during the luteal phase which may help to suppress FSH secretion or their actions.

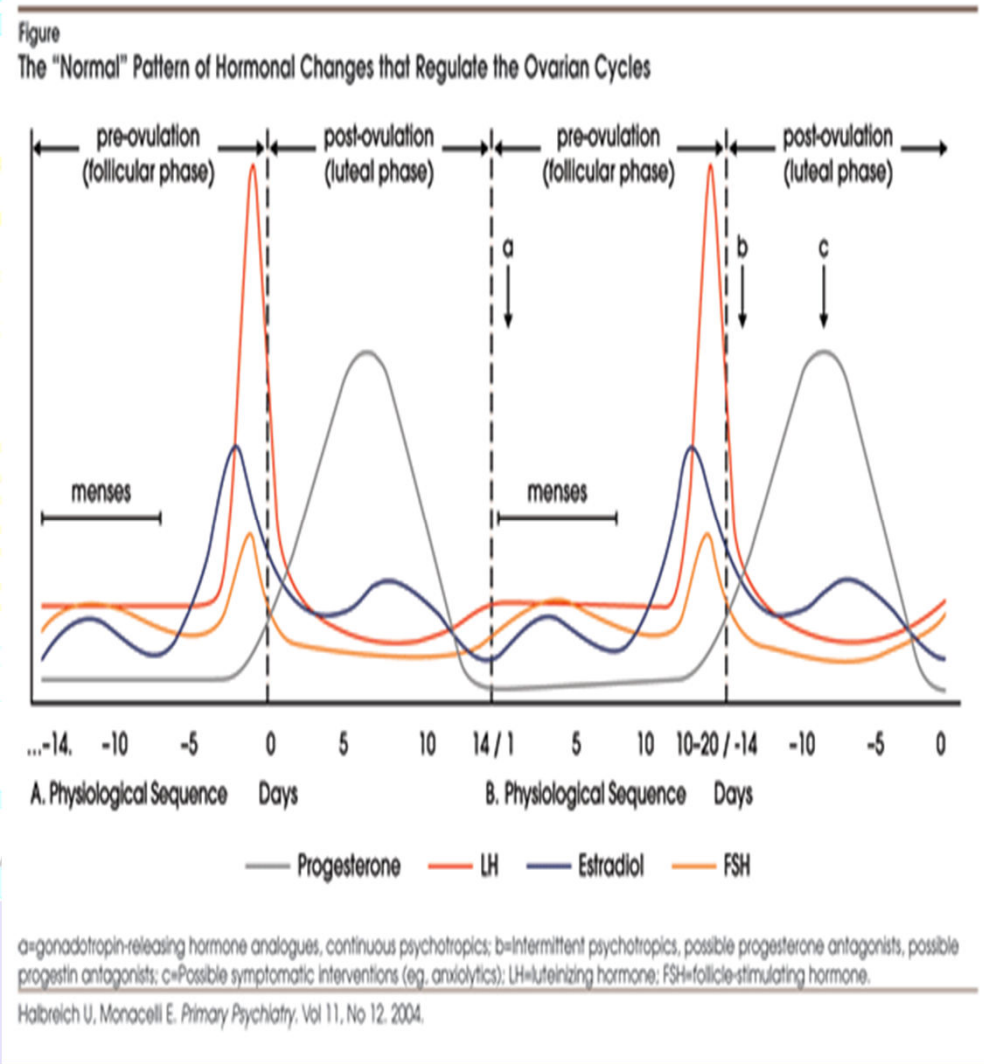
LUTEAL PHASE

- This retards the development of new follicles, so that further ovulation does not occur during that cycle.
- Multiple ovulations and possible pregnancies on succeeding days of cycle are prevented.



LUTEAL PHASE

- If the pregnancy does not occur than following sequence of events occur
- New follicles start to develop towards the end of one cycle in preparation for the next.
- Inhibin production is decreased at the end of luteal phase.



LUTEAL PHASE

- Estrogen and progesterone levels also fall during the late luteal phase (starting about day 22) because corpus luteum regresses and stops functioning.
- With the declining function of corpus luteum, esterogen and progesterone fall to very low levels by day 28 of the cycle.

Pregnancy Takes Place

- Corpus luteum persist up to four months now called as corpus luteum of pregnancy
- Becomes enlarge and secretes both the estrogens and progesterone
- Now the corpus luteum is under the control of developing placenta



REFERENCES.

- Keith L Moore Developing Human 8th edition
- Chapter 2
- Pages 23-24



THANK YOU

